# AJAY GANESAN IYER

Motion Planning | State Estimation | Robotics & Controls

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## EDUCATION

**Texas A&M University, College Station, TX** Aug 2024 - Present
Ph.D., Mechanical Engineering

**Indian Institute of Technology, Roorkee, India** Jun 2021
M.Tech., Systems and Control

**National Institute of Technology, Tiruchirappalli, India** May 2019
B.Tech., Instrumentation and Control

## RESEARCH EXPERIENCE

**Research Assistant, Texas A&M University** **Aug 2026 - Present**
*Frontier Mapping | Advisor: Dr. Srikanth Saripalli*

- Developing **vision-language model (VLM)-based semantic value mapping for frontier exploration** to guide further inspection when clutter, occlusion, and limited viewpoints make geometric coverage insufficient in unstructured environments.
- Formulating a **2D semantic value map** by projecting image-text relevance scores using depth and robot pose and fusing observations with visibility-based weights. Developing frontier ranking from map values, expected information gain, and travel cost to select inspection viewpoints.
- Building a **ROS 2/Gazebo Husky-Jackal framework** with LiDAR mapping, cameras, coordinate transforms, and extended Kalman filter localization to evaluate the proposed frontier selection under occlusion and restricted viewpoint access.

**Graduate Research Assistant, Texas A&M University** **Aug 2025 - Jul 2026**
*Autonomous Vehicles for All | Advisor: Dr. Reza Langari*

- Developed a **Frenet-frame motion planner** for marked and unmarked gravel roads, generating candidate trajectories from lateral offsets and velocity profiles and selecting feasible paths using curvature constraints and a weighted cost function.
- Supplied the planner with **road-boundary estimates from a real-time ROS perception pipeline**. Tuned CLRerNet lane detection and SAM drivable-area segmentation and aligned camera, LiDAR, and odometry data in a common map frame.
- Tested the integrated pipeline using **ROS bags and simulation** under low light, glare, dust, unmarked gravel, sensor dropout, and timing jitter. Addressed degraded trajectory candidates through perception-threshold tuning and temporal smoothing to maintain lane continuity.
- Studied **visual-token redundancy in OpenDriveVLA** to identify opportunities for efficient Vision-Language-Action (VLA) inference. Evaluated principal component analysis and token selection across **6,019 nuScenes validation samples** using UniAD and STP-3 planning metrics.
- Identified a compact representation of the visual inputs: approximately **10 principal components** captured **80% of scene-token variance** and **97% of map-token variance**, providing a basis for inference-time token reduction.

**Graduate Research Assistant, Texas A&M University** **Aug 2024 - Aug 2025**
*Automation in Welding | Advisor: Dr. Prabhakar Pagilla*

- Built a **Python/PyQt5/ROS interface** to automate **UR10e/Miller pulsed-MIG** weld setup and execution, integrating Arduino PWM/relay control for arc voltage, wire feed, and weld triggering.
- Registered CAD parts using **three torch-probed reference points, Kabsch alignment, and ICP refinement**. Generated STL-edge weld paths with surface-normal orientations and a 45° torch tilt without external metrology.
- Implemented **Newton-Raphson IK** and **RRT*** with wrist-limit handling and angle unwrapping. **Time-parameterized trajectories** for constant-tip-speed welding and coordinated weld triggering with approach/retreat motions. Added trajectory logging and joint-jump/jerk checks.
- Added **Optris PI 05M infrared weld-pool monitoring** to the execution pipeline, using bead-height and calibrated temperature feedback for closed-loop regulation of travel speed and arc voltage.

## TECHNICAL SKILLS

**Software:** Python, C++, C, Embedded C, MATLAB/Simulink, ROS 1/2, Git, Gazebo, RViz, CARLA, Webots
**Planning & Control:** Model Predictive Control (MPC), constrained optimization, RRT*, inverse kinematics, sliding mode control, PID, system identification, stability testing
**Estimation & Perception:** Kalman filtering, camera/LiDAR fusion, OpenCV, semantic segmentation, PCA, TensorFlow, Keras, Pandas
**Hardware & Interfaces:** UR10e, Arduino, Raspberry Pi, LabVIEW, PyQt5, infrared thermography
**Coursework:** Aerial Robotics, Control of Mobile Robots, Computational Motion Planning, Optimal Control, Spacecraft Dynamics and Control, Deep Learning

## INDUSTRY EXPERIENCE

**Robotics Engineer, MIKO** **Jun 2021 - Jun 2023**

*Mumbai, India*

- Developed **control and path-planning software for a head-motion manipulator**, using system identification, stability testing, and robust controller synthesis to produce expressive joint-level articulation.
- Extended the control work to a **self-balancing robot**, integrating the sensor suite and tuning robust controllers to maintain balance.
- Connected **audio direction-of-arrival estimates to yaw-axis control** so the robot could turn toward an active speaker in real time. Also delivered remote-control functionality for the mobile platform.

## RELATED PROJECTS

**LPV Model Predictive Control for Autonomous Vehicles** **Sep 2024 - Nov 2024**

*Texas A&M University*

- Developed **constrained model predictive control in Python** for vehicle trajectory tracking, using a six-state nonlinear bicycle model with steering and acceleration inputs and a **20 ms discretization step**.
- Updated the linearized dynamics along the prediction horizon and formulated a **10-step quadratic program solved with cvxopt**, combining tracking error, terminal error, and input-rate penalties through an augmented input-increment model.
- Constrained the optimization by **steering, speed, sideslip, and tire-force limits** ($\pm 30^\circ$ steering, 1-30 m/s longitudinal speed). Validated the resulting closed-loop controller on straight, sinusoidal, and circuit trajectories.

**Robust Nonlinear Control for Urban Vehicle Platooning** **Aug 2020 - Apr 2021**

*M.Tech. Thesis, IIT Roorkee*

- Designed longitudinal and lateral controllers for a **three-vehicle platoon** to maintain spacing and track a reference path at 8 m/s, using nonlinear bicycle dynamics in MATLAB/Simulink.
- Implemented and compared **PID, conventional sliding mode, power-rate exponential reaching law (PREL), and super-twisting control**, with a **Luenberger observer** to estimate yaw rate from lateral velocity.
- Assessed the controllers using tracking-error integrals and control energy across all three vehicles. **PREL achieved the lowest error indices**, while super-twisting provided lateral convergence without overshoot and the **lowest longitudinal control energy**.

**Kalman Filtering for Needle-Tip State Estimation** **Sep 2019 - Dec 2019**

*Indian Institute of Technology, Roorkee*

- Built a **LabVIEW needle-tip tracker** combining background subtraction, edge detection, ROI-based RANSAC line fitting, and a **four-state Kalman filter** to maintain position estimates when ultrasound speckle obscured the tip.
- Assessed tracking accuracy against an **NDI-Aurora electromagnetic tracker** in artificial and biological tissue at insertion depths up to 6.5 cm, using position root mean square error (RMSE).
- Compared with vision-only tracking, Kalman filtering reduced needle-tip position RMSE **from 0.79 cm to 0.03 cm in artificial tissue** and **from 0.84 cm to 0.21 cm in biological tissue**.

## RELEVANT INTERNSHIP

**Indian Institute of Technology, Bombay | EKF-SLAM** **Jun 2017 - Jul 2017**

*Mobile Robot Localization, Mapping and Navigation*

- Adapted a MATLAB **Extended Kalman Filter SLAM (EKF-SLAM)** simulator for joint robot-pose and landmark estimation under motion and sensor noise, using Jacobian linearization and Mahalanobis-gated nearest-neighbor data association.
- Observed reduced landmark uncertainty during repeated waypoint traversals. Extended the simulator with **reactive steering from three sensor views** to navigate around landmarks without preset waypoints.
- Implemented navigation on a physical **SPARK-V robot in Embedded C (ATmega16)**, combining ultrasonic obstacle avoidance with wheel-encoder distance feedback. Demonstrated collision-free traversal of a room containing obstacles.

## PUBLICATIONS & AWARDS

- A. G. Iyer, J. Samantaray, S. Ghosh, A. Dey, S. Chakrabarty. "Sliding Mode Control Using Power Rate Exponential Reaching Law for Urban Platooning." *7th International Conference on Advances in Control and Optimization of Dynamical Systems (ACODS)*, 2022.
- M. Adharsh, Ajay G. Iyer, Abhinav G. Athrey, Ashwin V., Venkatesh V. G. "Particle Swarm Optimization Algorithm for Finding the Brightest Spot in an Arena." *3rd IEEE International Conference on Electronics, Communication and Aerospace Technology*, 2019. **Best Paper Award**, 2019.